

1.0 OVERVIEW

The ENC28J60 is a stand-alone Ethernet controller with an industry standard Serial Peripheral Interface (SPI). It is designed to serve as an Ethernet network interface for any controller equipped with SPI.

The ENC28J60 meets all of the IEEE 802.3 specifications. It incorporates a number of packet filtering schemes to limit incoming packets. It also provides an internal DMA module for fast data throughput and hardware assisted checksum calculation, which is used in various network protocols. Communication with the host controller is implemented via an interrupt pin and the SPI, with clock rates of up to 20 MHz. Two dedicated pins are used for LED link and network activity indication.

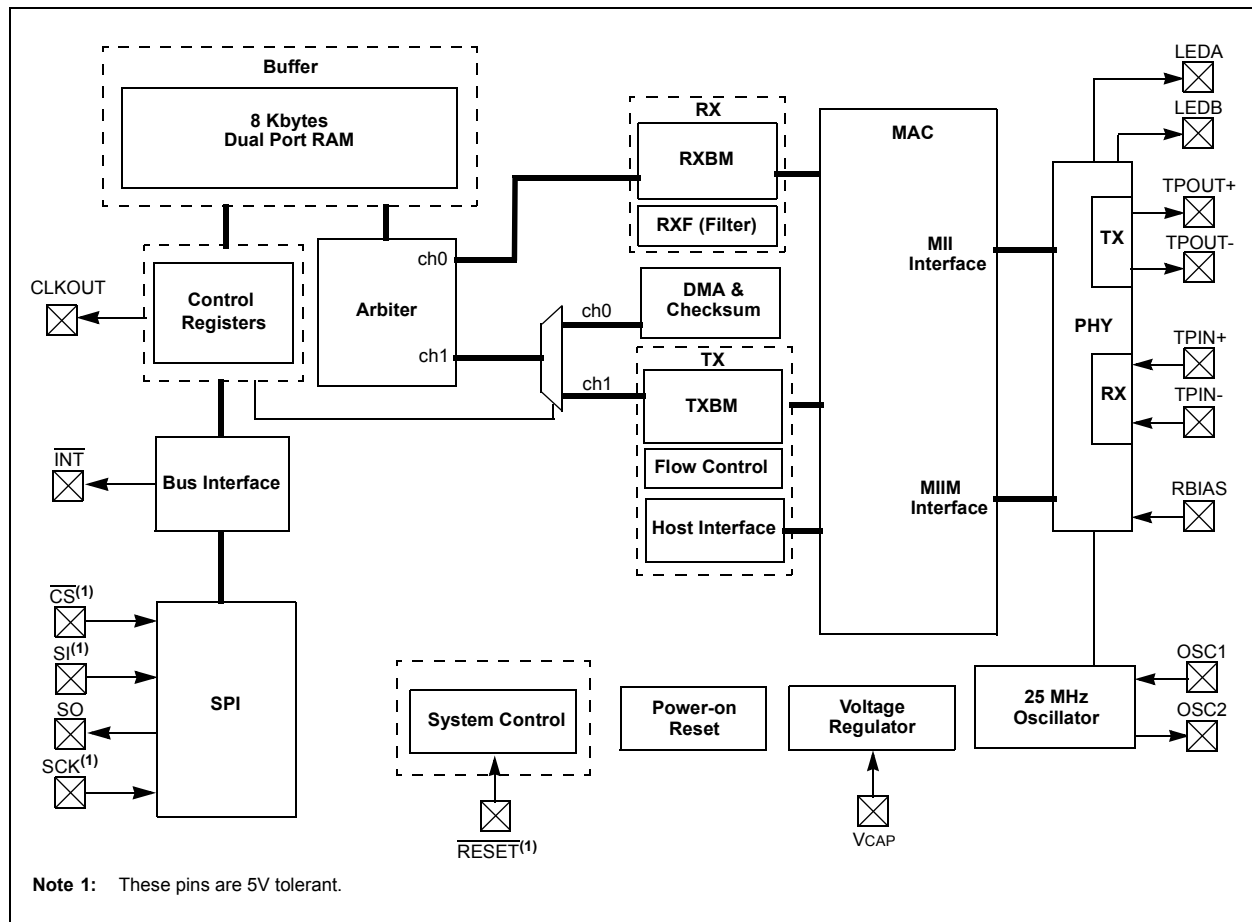
A simple block diagram of the ENC28J60 is shown in Figure 1-1. A typical application circuit using the device is shown in Figure 1-2. With the ENC28J60, two pulse transformers and a few passive components are all that are required to connect a microcontroller to an Ethernet network.

The ENC28J60 consists of seven major functional blocks:

1. An SPI interface that serves as a communication channel between the host controller and the ENC28J60.
2. Control registers which are used to control and monitor the ENC28J60.
3. A dual port RAM buffer for received and transmitted data packets.
4. An arbiter to control the access to the RAM buffer when requests are made from DMA, transmit and receive blocks.
5. The bus interface that interprets data and commands received via the SPI interface.
6. The MAC (Medium Access Control) module that implements IEEE 802.3 compliant MAC logic.
7. The PHY (Physical Layer) module that encodes and decodes the analog data that is present on the twisted-pair interface.

The device also contains other support blocks, such as the oscillator, on-chip voltage regulator, level translators to provide 5V tolerant I/Os and system control logic.

FIGURE 1-1: ENC28J60 BLOCK DIAGRAM



ENC28J60

FIGURE 1-2: TYPICAL ENC28J60-BASED INTERFACE

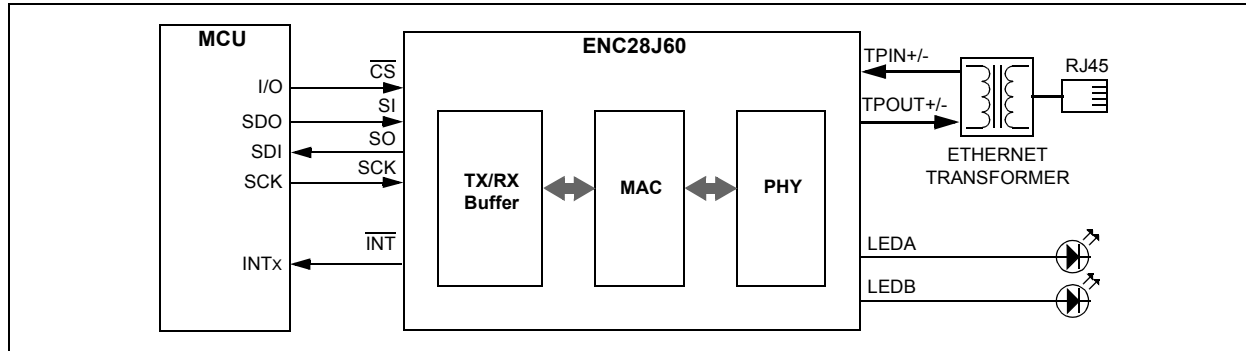


TABLE 1-1: PINOUT I/O DESCRIPTIONS

Pin Name	Pin Number		Pin Type	Buffer Type	Description
	SPDIP, SOIC, SSOP	QFN			
V _{CAP}	1	25	P	—	2.5V output from internal regulator. A low Equivalent Series Resistance (ESR) capacitor, with a typical value of 10 mF and a minimum value of 1 mF to ground, must be placed on this pin.
V _{SS}	2	26	P	—	Ground reference.
CLKOUT	3	27	O	—	Programmable clock output pin. ⁽¹⁾
INT	4	28	O	—	INT interrupt output pin. ⁽²⁾
NC	5	1	O	—	Reserved function; always leave unconnected.
SO	6	2	O	—	Data out pin for SPI interface. ⁽²⁾
SI	7	3	I	ST	Data in pin for SPI interface. ⁽³⁾
SCK	8	4	I	ST	Clock in pin for SPI interface. ⁽³⁾
CS	9	5	I	ST	Chip select input pin for SPI interface. ^(3,4)
RESET	10	6	I	ST	Active-low device Reset input. ^(3,4)
V _{SSRX}	11	7	P	—	Ground reference for PHY RX.
TPIN-	12	8	I	ANA	Differential signal input.
TPIN+	13	9	I	ANA	Differential signal input.
RBIAS	14	10	I	ANA	Bias current pin for PHY. Must be tied to ground via a resistor (refer to Section 2.4 “Magnetics, Termination and Other External Components” for details).
V _{DDTX}	15	11	P	—	Positive supply for PHY TX.
TPOUT-	16	12	O	—	Differential signal output.
TPOUT+	17	13	O	—	Differential signal output.
V _{SSTX}	18	14	P	—	Ground reference for PHY TX.
V _{DDRX}	19	15	P	—	Positive 3.3V supply for PHY RX.
V _{DDPLL}	20	16	P	—	Positive 3.3V supply for PHY PLL.
V _{SSPLL}	21	17	P	—	Ground reference for PHY PLL.
V _{SSOSC}	22	18	P	—	Ground reference for oscillator.
OSC1	23	19	I	ANA	Oscillator input.
OSC2	24	20	O	—	Oscillator output.
V _{DDOSC}	25	21	P	—	Positive 3.3V supply for oscillator.
LEDB	26	22	O	—	LEDB driver pin. ⁽⁵⁾
LEDA	27	23	O	—	LEDA driver pin. ⁽⁵⁾
V _{DD}	28	24	P	—	Positive 3.3V supply.

Legend: I = Input, O = Output, P = Power, ANA = Analog Signal Input, ST = Schmitt Trigger

Note 1: Pins have a maximum current capacity of 8 mA.

2: Pins have a maximum current capacity of 4 mA.

3: Pins are 5V tolerant.

4: Pins have an internal weak pull-up to V_{DD}.

5: Pins have a maximum current capacity of 12 mA.